Verified Symmetry Hypotheses and Causal Repair in LLM-Guided Quantum Circuit Search

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Abstract

We study how large language models (LLMs) turn *symmetry hypotheses* into *executable* quantum models, and whether an executable verifier and an analytic repair step *causally* improve that process under matched evaluation budgets. The mathematical tools we use—group twirls, channel and predictor equivariance residuals, and an orthogonal-projection risk identity—are established; our object of study is behavioural: the gap between a model *proposing* a correct transformation, *implementing* it correctly, and *obtaining useful task performance*. We make three contributions. First, we audit a prior LLM-evolutionary quantum-ML study and show that its model selection leaked test data, that a central architecture-wide equivariance claim is false (measured channel defect 0.0597), and that its significance was computed at the proposal rather than run level. Second, we introduce a confound-free causal-repair experiment with an analytically known invariant-risk floor and a *gate-count-matched* asymmetric control; on a ring benchmark we show, with 24 paired draws and bootstrap confidence intervals, that enforcing equivariance *causally improves* held-out risk exactly when the target is symmetric and *causally hurts* when it is not, with the effect attributable to symmetry rather than capacity. Third, a matched-budget, single-model, five-arm search comparison (120 runs, 12 paired seeds) yields a clean *null* on the headline hypothesis that verifier feedback improves held-out risk, while showing that LLM search beats non-LLM baselines only when the task is not dominated by a trivial shortcut. We report negative and null results as first-class findings and scope every claim to the paired evidence.

1 Introduction

Symmetry is a first-class inductive bias in quantum machine learning (QML): a circuit that commutes with a representation of the data's symmetry group, combined with a compatible readout, yields invariant predictions. A tempting hypothesis is that an LLM asked to design such circuits will both *state* the right symmetry and *build* it. Prior work has reported LLM-driven evolution of symmetry-aware quantum circuits, but the strength of that evidence had not been established.

This paper treats the hypothesis as measurable rather than anecdotal, and separates three levels that casual "symmetry-aware LLM" claims conflate: (i) **proposing** a transformation (a stated group hypothesis); (ii) **implementing** it (the circuit actually commutes with the representation); and (iii) **performing** (the trained predictor generalises on held-out data). Our thesis is that the interesting, testable phenomenon is the *gap* between these levels and whether executable feedback closes it.

Contributions.

1. **A reproducibility audit** (§4) of a prior ShinkaEvolve-based study: test-set leakage in selection, a refuted architecture-wide equivariance claim, and a proposal-level (not run-level) significance error. These bound what the old numbers can mean.
2. **A confound-free causal-repair experiment** (§5.1, §6.1) with targets $y_\eta = h + \eta b$, an analytic invariant-risk floor $\eta^2 \mathbb{E}[b^2]$, and a gate-count-matched asymmetric control that isolates the symmetry effect from expressivity. We show repair *helps iff the target is invariant*.

3. **A matched-budget search study** (§5.2, §6.2) comparing random, mutation-only, LLM-fitness, LLM-verify, and LLM-verify-repair arms with identical initial populations, every invalid proposal retained, and a sealed-holdout guard. We report a clean **null** on verifier feedback and a task-dependent LLM advantage.

All results are prospective and pre-declared, with run-level paired uncertainty; negative and null results are reported as such and no claim outruns the paired evidence.

2 Related work

Representation-based gate symmetrisation and invariant re-uploading appear in Meyer et al. [1]; constructive equivariant unitary layers and channels, including continuous groups, in Nguyen et al. [2]. Permutation-invariant QML for graph properties including approximate symmetry is studied by Mansky et al. [3]; Baumann and Linnhoff-Popien [4] show that symmetry alone does not determine a useful architecture. Averaging-operator generalisation theory appears in Elesedy and Zaidi [5]; Lie-derivative symmetry promotion in Otto et al. [6]; finite-measurement task-level symmetry inference, task validation and projection bias in Chen [7]. ShinkaEvolve's parent sampling, novelty rejection and bandit model selection appear in Lange et al. [8]. The mathematical tools are therefore not the novelty; the behavioural, intervention-based study of the propose→verify→implement→perform gap is the candidate contribution, and priority still requires a final broader review.

3 Preliminaries: continuous symmetry diagnostics

Let P_g be a specified unitary representation of a finite group G on $\mathbb{C}^d$ and U a circuit unitary at fixed numerical parameters. The *strict* residual

$$D_U(g) \;=\; \frac{\|U - P_g U P_g^\dagger\|_F^2}{2d} \in [0,2] \tag{1}$$

vanishes iff U commutes with P_g. For physical unitary channels we use the *channel* residual

$$D_{\text{chan}}(g) \;=\; 1 - \left|\mathrm{Tr}(U^\dagger P_g U P_g^\dagger)/d\right|^2 \in [0,1], \tag{2}$$

which vanishes exactly when conjugation differs by a global phase (e.g. Z anticommutes with X: strict residual 2, channel residual 0). Adjacent transpositions generate S_n; zero residual on all generators certifies exact symmetry at those parameters. A random-parameter zero is evidence, not an architecture-wide proof.

The Reynolds operator $\mathcal{T}_G(U) = \frac{1}{|G|}\sum_g P_g U P_g^\dagger$ is an orthogonal projection, so the mean strict residual over a uniform group equals $\|U - \mathcal{T}_G(U)\|_F^2/d$. For predictions f on complete orbits, $D_f = \mathbb{E}_x \,\mathrm{Var}_g\, f(gx)$, and under a group-invariant input measure with an invariant target the squared-risk projection identity

$$R(f) \;=\; R(\mathcal{T}_G f) + \mathbb{E}\big[(f - \mathcal{T}_G f)^2\big] \tag{3}$$

holds by orthogonality of the projection. Constant predictors are perfectly invariant, so predictive risk and output variance are always reported alongside invariance. Circuit equivariance is sufficient for output invariance only with a compatible readout; it is not necessary.

4 A reproducibility audit of prior LLM-evolutionary QML

We re-examined a prior study that evolved quantum circuits with an adaptive ensemble of commercial frontier models (weak/mid/frontier tiers, UCB bandit selection) on a graph-connectedness task. Four findings bound its claims.

Test leakage in selection. The selection score used a train–*test* gap and the public feedback disclosed test accuracy and loss. The historical "test" set therefore measured an adaptively reused development set and cannot establish held-out generalisation.

A refuted equivariance claim. The hand-designed baseline was claimed to be permutation-equivariant for all parameters. Its one-direction CRZ layer is not: at a documented parameter setting its mean adjacent-generator channel defect is 0.0597, and a bidirectional half-angle repair reduces it to 1.5×10^{-14} (floating-point roundoff). A separate ordered-CRX counterexample shows that covering every unordered pair does not guarantee symmetry, because non-commuting controlled rotations are order dependent.

Non-independence. 563 selected proposals are not 563 independent experiments: they share ancestry, data and adaptive selection, yet significance was computed proposal-by-proposal, inflating confidence.

A trivial shortcut. A single edge-count threshold fitted on training data alone attains 83.0% validation accuracy (predict connected iff ≥ 9 edges), indicating that density, not structure, carries much of the signal; no quantum advantage follows from that task.

5 Methods

5.1 Confound-free causal repair (WP-REPAIR)

Targets. On the ring benchmark (inputs $x \in \mathbb{R}^6$, group D_6 of order 12 acting by site relabelling), let h be the dataset's ground-truth *invariant* oracle $h(x) = \frac{1}{n}\sum_i x_i x_{i+1}$ and let a frozen random direction v induce a breaking component $b(x) \propto \big(v{\cdot}x - \mathcal{T}_G(v{\cdot}\cdot)\big)$, i.e. the group-centred remainder of a linear functional, normalised so $\mathbb{E}_{\text{train}}[b^2] = 1$. Targets are

$$y_\eta(x) \;=\; h(x) + \eta\, b(x), \qquad \eta \in \{0, 0.5, 1\}. \tag{4}$$

Crucially, h and b are functions of the *input*, never produced by any compared ansatz, removing the confound in which targets are generated by the architectures being compared. The Reynolds projection is computed by applying the *explicit* group closure to each input (the dataset's orbit labels are singletons and cannot be used).

Invariant-risk floor. Because h is invariant and b is group-centred, no invariant predictor can beat $\text{floor}(\eta) = \eta^2\,\mathbb{E}[b^2]$, and the empirical orbit-mean floor equals it (verified to 10^{-6}).

Matched arms. Three ansatz arms share parameter count $\{a, b, c\}$ and a **fixed symmetric readout** (equal weight on every Z_i), so an equivariant block yields an invariant predictor:

- `original`: one-direction CRZ layer, 15 CRZ gates (asymmetric);
- `repaired`: bidirectional shared-angle CRZ, 30 CRZ gates (exactly equivariant, measured channel residual 0);
- `control`: repeated same-direction CRZ, 30 CRZ gates (asymmetric, operator-family-equivalent to `original`, with the *same gate count as* `repaired`).

Only the three block parameters are trained (Adam, MSE, fixed budget, common random numbers: a seed gives an identical initialisation across arms). Selection is on the training split only; development risk is reported per (teacher, seed, η, arm); the locked holdout is never read. The decisive contrast is `repaired` vs `control`: they differ *only* in symmetry.

5.2 Matched-budget search (WP-SEARCH)

Five arms run at an identical evaluated-candidate budget B=12 with identical seed-derived initial populations: `random`, `mutation-ea`, `llm-fitness` (scalar fitness), `llm-verify` (adds an executable equivariance residual and a label-preservation report for the proposal's own hypothesis), and `llm-verify-repair` (compiles an inference-time Reynolds average when the hypothesis passes label preservation and the residual exceeds a tolerance). Every invalid proposal is retained in the denominator; per-run cost (tokens, evaluator calls, wall time) is logged. Model identity and reasoning effort are fixed categorical treatments (no adaptive bandit), so capability is not confounded with cost or exposure. The primary endpoint is the paired run-level difference in development-selected risk between `llm-verify` and `llm-fitness`; secondary contrasts are Holm-corrected within family. Development selection only; the locked holdout is never read by search.

Table 1: WP-REPAIR paired development-risk differences (a − b; negative ⇒ a better), n=24. Bootstrap 95% CIs. The invariant-risk floor is $\eta^2\mathbb{E}[b^2]$.

η	floor	contrast	mean diff	95% CI	frac. better
0.0	0.000	repaired vs control	**−0.00093**	[−0.00160, −0.00036]	0.62
0.0	0.000	repaired vs original	−0.00201	[−0.00308, −0.00095]	0.88
0.0	0.000	control vs original	−0.00107	[−0.00203, −0.00017]	0.62
0.5	0.217	repaired vs control	+0.01817	[+0.01320, +0.02307]	0.17
0.5	0.217	repaired vs original	+0.01690	[+0.01259, +0.02112]	0.17
0.5	0.217	control vs original	−0.00127	[−0.00544, +0.00286]	0.46
1.0	0.866	repaired vs control	**+0.04523**	[+0.03231, +0.05758]	0.04
1.0	0.866	repaired vs original	+0.04319	[+0.03493, +0.05139]	0.00
1.0	0.866	control vs original	−0.00204	[−0.01338, +0.00898]	0.42

Table 2: WP-REPAIR mean development risk and measured channel residual per arm (n=24). The repaired arm is exactly equivariant (residual 0); the asymmetric arms are not.

		`original`	`repaired`	`control`
η=0.0	dev risk	0.0151	**0.0131**	0.0140
η=0.0	channel res.	0.561	**0.000**	0.732
η=1.0	dev risk	0.9280	0.9711	**0.9259**
η=1.0	channel res.	0.757	**0.000**	0.752

Statistics. Contrasts are paired by seed (and, for LLM-vs-LLM, by model treatment). We report mean paired differences with 10,000-resample bootstrap 95% confidence intervals at the run level and the fraction of pairs improved; secondary families are Holm-corrected.

6 Results

6.1 Causal repair (WP-REPAIR)

Table 1 reports mean paired development-risk differences (a−b; negative means a is better) over 24 paired draws (3 frozen teachers × 8 optimiser seeds) on `ring_n6`. Table 2 reports mean development risk and the measured channel residual per arm.

Finding. When the target is invariant (η=0), enforcing equivariance *causally improves* held-out risk, and it does so against a control matched in gate count (CI excludes 0), so the benefit is attributable to symmetry rather than capacity. When the target breaks symmetry (η=1), enforcing equivariance *causally hurts*: the repaired arm is pinned near the analytic invariant floor 0.866 and cannot represent the breaking component, whereas the asymmetric arms fit it. The control tracks the original at every η (CI includes 0), validating that it isolates gate count from symmetry.

6.2 Matched-budget search (WP-SEARCH)

Table 3 reports mean development-selected risk per arm; Table 4 reports paired contrasts (12 paired seeds per family, one model treatment `ycrc-local`/none, 120 runs total).

Findings. The headline hypothesis is a clean **null**: verifier feedback does not significantly change development-selected risk on either family (CIs include 0). This is the protocol's anticipated "changes the measured residual but not the predictive risk" branch—an implementation-reliability result, not a performance win. LLM search beats non-LLM baselines on `ring_n6` (Holm $p < 0.05$) but not on `graphs_n6`,

Table 3: WP-SEARCH mean development-selected risk by arm (n=12 runs each, B=12). On `graphs_n6` a density shortcut dominates and no arm separates; on `ring_n6` LLM arms beat non-LLM baselines.

arm	`ring_n6` dev risk	`graphs_n6` dev risk
`random`	0.01013	0.53782
`mutation-ea`	0.01022	0.53930
`llm-fitness`	**0.00651**	0.56572
`llm-verify`	0.00709	0.55158
`llm-verify-repair`	0.00664	0.55250

Table 4: WP-SEARCH paired contrasts (`a` − `b`; negative ⇒ `a` better), n=12. Primary: verifier vs fitness. Secondary Holm-adjusted within family; * denotes $p_{holm} < 0.05$.

family	contrast	mean diff	95% CI	note
graphs	`verify` − `fitness`	−0.0141	[−0.0591, +0.0284]	null
ring	`verify` − `fitness`	+0.0006	[−0.0003, +0.0014]	null
ring	`fitness` − `mutation-ea`	−0.0037	[−0.0042, −0.0033]	*
ring	`verify` − `random`	−0.0030	[−0.0039, −0.0021]	*
ring	`verify-repair` − `verify`	−0.0005	[−0.0013, +0.0005]	null
graphs	`fitness` − `mutation-ea`	+0.0264	[−0.0172, +0.0644]	null

where a density shortcut dominates. An inference-time Reynolds-averaging benefit visible in a 3-seed pilot did *not* survive to 12 seeds ($p_{holm} = 0.61$).

7 Discussion and conclusion

The two studies are consistent and jointly informative. Symmetry is a real, causally verifiable inductive bias *when the target is invariant* (WP-REPAIR, gate-count-controlled, with an analytic floor as reference), but bolting a verifier onto fixed-budget LLM search did not, by itself, improve held-out risk here (WP-SEARCH null), and any LLM advantage is task-dependent and vanishes under a strong shortcut. This supports a narrower, defensible claim than the prior study's: the value of verification in this setting is *implementation reliability*—detecting and repairing a wrong symmetry— rather than a uniform performance lift, and the benefit of symmetry itself is conditional on the task actually being symmetric.

Limitations. Small dense statevector simulation; no quantum-advantage claim (a trivial edge-count baseline is strong on the historical task); a single local model family, so results are not numerically comparable to the prior commercial-ensemble study; a fixed budget B=12; and one held-out task family reserved for transfer rather than a claim of open-ended discovery. Confirmatory transfer on the held-out family, larger budgets, and additional model treatments are future work.

Reproducibility and integrity. All splits are orbit-disjoint; locked holdouts were never read by search and are evaluated at most once; every result JSON records seeds, module versions, git commit and host; analysis is pre-declared with run-level paired bootstrap confidence intervals and Holm correction. Code, data manifests and run logs accompany the paper.

References

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Appendix: reproduction

Core and experiment tests:

```
python3 -m unittest discover -s research/core      -p 'test_*.py'   # 36 pass
python3 -m unittest research.experiments.repair.test_repair          # 6 pass
python3 -m unittest discover -s research/search     -p 'test_*.py'   # 14 pass
python3 -m research.experiments.repair.repair_experiment \
  --teacher-seeds 101,202,303 --n-opt-seeds 8 --budgets 200 \
  --max-train 256 --out research/experiments/repair/repair_results.json
python3 -m research.search.analyze --dir logs/runs_confirm \
  --out research/search/confirm_analysis.json
python3 -m research.compile_results \
  --repair research/experiments/repair/repair_results.json \
  --search research/search/confirm_analysis.json \
  --out research/results_summary.md
```

Stack: `Python/3.12.3`, `SciPy-bundle/2024.05` (NumPy 1.26.4, SciPy 1.13.1); LLM arms additionally load the YCRC-hosted model module (`ycrc-local`). CPU batch jobs run on the cluster `day` partition; `OPENBLAS_NUM_THREADS=1`.